

Task 2: Dataset Preparation

Dataset Used: Iris Dataset

The Iris dataset is a commonly used dataset in Machine Learning. It contains information about different species of iris flowers based on their sepal and petal measurements. The dataset contains **150 records and 6 columns**, including an ID column and the target variable, Species.

1. Loading the Dataset

The Iris dataset was loaded using Pandas. Pandas is a Python library that provides useful tools for handling and analyzing structured data.

After loading the dataset, it was stored in a DataFrame so that the data could be easily inspected and processed.

```
[1]:
import pandas as pd

df = pd.read_csv("Iris.csv")

print("Dataset loaded successfully!")

Dataset loaded successfully!

[2]:
```

2. Displaying the First 5 Rows

The first five rows of the dataset were displayed using the head() function. This helps us understand the structure of the dataset and identify the available columns.

The dataset contains the following columns:

- **Id**
- **SepalLengthCm**
- **SepalWidthCm**
- **PetalLengthCm**
- **PetalWidthCm**
- **Species**

The first four measurements describe the physical characteristics of the flower, while Species represents the type of iris flower.

[2]:

```
df.head()
```

[2]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

3. Checking the Shape of the Dataset

The shape of the dataset was checked to determine the number of rows and columns.

The result was **150 rows and 6 columns**. This means the dataset contains 150 flower records and 6 attributes.

[3]:

```
print("Shape of the dataset:", df.shape)
```

Shape of the dataset: (150, 6)

4. Checking Data Types

The data types of all columns were checked to understand how the information is stored.

The measurement columns such as Sepal Length, Sepal Width, Petal Length, and Petal Width contain numerical values. The Species column contains categorical text values representing the type of iris flower.

The Id column is also numerical, but it is only used as an identifier and is not useful as a Machine Learning feature.

[4]:

```
print("Data types of each column:")
print(df.dtypes)
```

```
Data types of each column:
Id                int64
SepalLengthCm    float64
SepalWidthCm     float64
PetalLengthCm    float64
PetalWidthCm     float64
Species          object
dtype: object
```

5. Identifying Missing Values

Missing values were checked in each column using Pandas.

Since the original Iris dataset does not normally contain missing values, a few missing values were temporarily introduced for demonstration purposes. This allowed the data-cleaning process to be performed and observed.

The missing-value check showed one missing value in each of the four measurement columns where the values were intentionally removed.

[5]:

```
# Create a copy so the original dataset remains unchanged
df_missing = df.copy()

# Create some missing values for demonstration
df_missing.loc[2, "SepalLengthCm"] = None
df_missing.loc[10, "SepalWidthCm"] = None
df_missing.loc[20, "PetalLengthCm"] = None
df_missing.loc[30, "PetalWidthCm"] = None

print("Missing values in each column:")
print(df_missing.isnull().sum())
```

```
Missing values in each column:
Id                0
SepalLengthCm    1
SepalWidthCm     1
PetalLengthCm    1
PetalWidthCm     1
Species          0
dtype: int64
```

6. Handling Missing Values

The missing numerical values were handled by replacing them with the **mean value of their respective columns**.

Using the mean is one common method for handling missing numerical data. It allows us to keep the existing records instead of deleting rows from the dataset.

After filling the missing values, the dataset was checked again. The result showed **zero missing values** in all columns.

[6]:

```
# Fill missing numerical values with the mean of their respective columns

numerical_columns = [
    "SepalLengthCm",
    "SepalWidthCm",
    "PetalLengthCm",
    "PetalWidthCm"
]

for column in numerical_columns:
    df_missing[column] = df_missing[column].fillna(df_missing[column].mean())

print("Missing values after cleaning:")
print(df_missing.isnull().sum())
```

```
Missing values after cleaning:
Id                0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

7. Identifying Features and Target

For this Machine Learning task, the four flower measurements were selected as the **features**, while Species was selected as the **target**.

Features

- SepalLengthCm
- SepalWidthCm
- PetalLengthCm
- PetalWidthCm

Target

- Species

The **Id** column was not selected as a feature because it is simply an identification number and does not describe any characteristic of the flower.

[7]:

```
features = [  
    "SepalLengthCm",  
    "SepalWidthCm",  
    "PetalLengthCm",  
    "PetalWidthCm"  
]
```

```
target = "Species"
```

```
print("Features:")  
print(features)
```

```
print("\nTarget:")  
print(target)
```

Features:

['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']

Target:

Species

8. Separating the Dataset into X and y

The dataset was separated into two parts:

X (Features): Contains the four numerical flower measurements.

y (Target): Contains the Species column.

This separation is important because Machine Learning models use **X as input data** and learn to predict **y as the output**.

For example, the model can use sepal and petal measurements in X to predict whether a flower belongs to Iris-setosa, Iris-versicolor, or Iris-virginica.

[8]:

```
X = df[features]
y = df[target]

print("Features (X):")
print(X.head())

print("\nTarget (y):")
print(y.head())
```

Features (X):

	SepallLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

Target (y):

0	Iris-setosa
1	Iris-setosa
2	Iris-setosa
3	Iris-setosa
4	Iris-setosa

Name: Species, dtype: object

9. Final Dataset Summary

After completing the preparation and cleaning process, the final dataset was checked.

The final dataset contains:

- **150 rows**
- **6 columns**
- **4 Machine Learning features**
- **1 target variable**
- **0 missing values**

The dataset is now prepared and can be used for further Machine Learning tasks such as classification.

[9]:

```
print("Dataset Preparation Summary")
print("-----")
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
print("Number of features:", X.shape[1])
print("Target variable:", target)
print("Total missing values:", df.isnull().sum().sum())
```

Dataset Preparation Summary

Number of rows: 150

Number of columns: 6

Number of features: 4

Target variable: Species

Total missing values: 0

10. Dataset Preparation Process

The overall process followed in this task was:

Load Dataset → Inspect Data → Check Shape → Check Data Types → Identify Missing Values → Handle Missing Values → Identify Features & Target → Separate X and y

This process helps ensure that the dataset is properly organized and ready for use in a Machine Learning model.

Conclusion

In this task, the Iris dataset was successfully loaded and prepared using Pandas. The dataset was inspected by checking its first few records, shape, and data types. Missing values were intentionally introduced to demonstrate a practical data-cleaning process and were then filled using the mean of the corresponding numerical columns.

The four flower measurements were selected as features, while Species was selected as the target variable. Finally, the dataset was divided into **X (features)** and **y (target)**, making it ready for further Machine Learning analysis and classification.